Semih Caglar

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EDUCATION

Bilkent University **Sep 2023 – Expected Jun 2028**
B.Sc. in Computer Science (Full Scholarship) *Ankara, Türkiye*

- **GPA: 3.96/4.00**
- **Ranked 1st** out of 169 students in the department
- Minor degree in **Psychology**

EXPERIENCE

Research Intern **Jun 2026 – Present**
SWAT Lab, Polytechnique Montréal *Montréal, Canada*

- Conducting research with Prof. Foutse Khomh in **LLM-based software engineering** and developer decision-making.

Undergraduate Researcher **Jan 2025 – Present**
Bilsen - Bilkent Software Engineering Research Group *Ankara, Türkiye*

- Conducting research with Prof. Eray Tuzun in **developer productivity** and automated software engineering.
- Co-authored **2 conference papers** and **2 workshop papers** in top venues.
- Built and deployed **SmellSolver** — a GitHub/Azure DevOps PR bot (**Python, Flask, SQLite**) that detects and repairs inline comment smells, monitoring over **50 Azure Repos** at **Beko**.
- Reviewed **several papers** in top venues.

Robotics Software Engineer Intern **Jun 2025 – Aug 2025**
Kuartis *Ankara, Türkiye*

- Built a ROS2 path-planning module that combines **Hybrid A*** algorithm with an Artificial Potential Field (**APF**) post-optimizer to generate non-holonomic paths for autonomous surface vessels.
- Implemented the constrained APF optimizer in modern **C++** with **NLopt**, enforcing collision-free corridor and curvature constraints.
- Validated the system in 1 km^2 simulated maritime environments, achieving 20 m path precision with optimization completing in 1 second per run.

Machine Learning Engineer **Apr 2024 – Sep 2024**
IDA AI *Remote*

- Developed and deployed machine learning algorithms for innovative fashion tech solutions, including **text2image garment design** and **virtual try-on**.
- Contributed to ModaMind's fashion design application and infrastructure using **Google Cloud, Firebase, Docker, and Kubernetes**.

PUBLICATIONS

- K. K. Dikici, S. Kara, **S. Çağlar**, E. Tüzün, and S. Sav. "SERSEM: Selective Entropy-Weighted Scoring for Membership Inference in Code Language Models." *FSE/AIware 2026 Poisoned Chalice Competition*, **1st Place**, 2026. doi:10.1145/3803437.3807734.
- **S. Çağlar**, Ş. E. Gökırmak, and E. Tüzün. "Automated Classification of Human Code Review Comments with Large Language Models." *EASE 2026* **(CORE A)**, 2026. arXiv:2604.23667.
- H. K. Çağlar, **S. Çağlar**, and E. Tüzün. "Automated Inline Comment Smell Detection and Repair with Large Language Models." *IEEE/ACM ASE 2025* **(CORE A*)**, pp. 1008–1019. doi:10.1109/ASE63991.2025.00088.
- V. Karakaya*, **S. Çağlar***, Y. Y. Korkmaz*, and E. Tüzün. "ATIBA: Grounded Integrity and Quality Checking for Research Papers." *International Workshop on Automated Techniques for Integrity and Quality in Software Engineering Research (ATIQSER 26)*, 2026. (to appear)

AWARDS & ACTIVITIES

Social Relations Coordinator – ACM Bilkent Student Chapter **2025 – Present**

Honorable Mention – ICPC Southeastern Europe Regional Contest (Bucharest, Romania) **2024**

Valedictorian – Ankara Science High School **2023**

Bronze Medal – National Olympiads in Informatics **2022, 2023**

Official Participant – Asia-Pacific Informatics Olympiad **2022**

PROJECTS

Deep Review (Academic paper review management system) | *Next.js, Express, TypeScript, TypeORM, SQLite*

- Designed and implemented an **LLM-powered review assistant** featuring **AI-generated reviews**, **PDF annotation**, **empirical checklist generation**, **compliance analysis**, and **reference verification**.
- Built **AI backend services** and integrated **LLM-powered review workflows** into paper management and **multi-round peer review pipelines**.

SplitMate (Expense sharing desktop app) | *JavaFX, Java Spring Boot, MongoDB*

- Led a team of **5 people** and owned the backend, designing Spring Boot services, defining MongoDB schemas for users, groups, transactions, debts, and integrating the desktop UI with backend APIs.
- Built the **automatic debt settlement** pipeline using **Python 3** and **PuLP** for integer linear programming.